Corporate Venture Investment Intelligence & Target Selection

Living Project Documentation — Version 1.5

Status: Financial screening + SQL analysis completed; Power BI dashboard in progress (Executive Overview built, Performance Comparison in build)

1. Document Purpose

This document is the living project playbook for the Corporate Venture Investment Intelligence & Target Selection portfolio project. It records the business objective, data sources, ETL methodology, financial analysis, investment-intelligence logic, SQL analysis, validation, errors encountered, current conclusions, and the next Power BI stage.

The Jupyter Notebook contains the implementation details. This document captures the professional project story, decisions, evidence, and interview-ready explanations.

2. Business Problem

Corporate venture and strategic investment teams may need to screen many public companies before allocating detailed due-diligence resources. This project builds a data-driven screening workflow that consolidates public financial information, evaluates financial performance and direction, identifies potential warning signals, and prioritizes companies for further human review.

The solution is a decision-support screening framework, not an autonomous investment adviser.

3. Project Objectives

- Build a reliable, source-backed financial dataset.
- Automate public-company financial extraction using SEC EDGAR/XBRL data.
- Create reusable Python ETL rather than relying on manual extraction.
- Standardize company-year financial information.
- Calculate growth, profitability, cash-generation and directional indicators.
- Create transparent investment-intelligence flags.
- Persist the analytical output in MySQL.
- Use SQL to compare, rank and classify potential targets.
- Present the results through a professional Power BI dashboard.

4. Company Universe

Company	Ticker	Financial Period
Etsy, Inc.	ETSY	FY2023–FY2025
Shake Shack Inc.	SHAK	FY2023–FY2025
Crocs, Inc.	CROX	FY2023–FY2025
Boot Barn Holdings, Inc.	BOOT	FY2023–FY2025
Duolingo, Inc.	DUOL	FY2023–FY2025

5. Project Roadmap & Status

Phase	Scope	Status
Phase 1	SEC/XBRL data acquisition	Completed
Phase 2	Reusable financial ETL	Completed
Phase 3	Financial analytical dataset	Completed
Phase 4	Investment intelligence	Completed
Phase 5	SQL screening & candidate classification	Completed
Phase 6	Broader investment insights	Planned
Phase 7	Power BI dashboard	In progress
Phase 8	Portfolio packaging & interview story	Planned

6. Data Sources & Technology Stack

- SEC EDGAR / data.sec.gov Company Facts XBRL API — primary financial source.
- Python — requests and pandas for extraction, transformation, QA and analysis.
- Jupyter Notebook / Anaconda — development environment.
- MySQL / MySQL Workbench — persistent analytical storage and SQL analysis.
- PyMySQL + SQLAlchemy — Python-to-MySQL connectivity.
- Power BI Desktop — visualization and decision-support layer, connected to MySQL; DAX measures, calculated columns and calculated tables.
- HTML prototypes — three layout prototypes used to agree the dashboard design before building it in Power BI.

7. Financial ETL Methodology

- Configure company identifiers, fiscal years and required XBRL concepts.
- Retrieve SEC Company Facts JSON.
- Identify appropriate US-GAAP concepts and USD units.
- Filter annual observations for FY2023–FY2025.
- Resolve repeated comparative observations and retain the appropriate filing record.
- Standardize monetary values to USD thousands.
- Build a consistent company-year financial structure.
- Calculate growth, margins, changes and directional signals.
- Run QA checks for year coverage, duplicates, nulls and structure.
- Build investment-intelligence flags.
- Load the final company-level screening output into MySQL.

8. Key Financial Metrics

The financial foundation uses revenue, gross profit where available, operating income, net income and operating cash flow. Derived measures include revenue growth, gross margin, operating margin, net margin, CFO margin, changes in margins, CFO-to-net-income relationships, and operating-income change.

9. Cross-Company Financial Results

Company	Revenue Growth 23–25	Operating Margin	Net Margin	CFO Margin	Operating Income Change

					($M)
DUOL	95.36%	13.07%	39.91%	37.38%	+148.83
SHAK	32.90%	4.32%	3.16%	15.38%	+56.59
BOOT	15.29%	12.52%	9.47%	7.72%	+7.57
ETSY	4.92%	9.23%	5.65%	24.05%	-13.63
CROX	1.98%	3.70%	-2.01%	17.58%	-887.27

10. Directional Investment-Intelligence Framework

The screening framework evaluates five directional signals: growth, operating margin, net margin, cash generation and operating income. Each company receives positive and negative direction counts. The net-direction score is calculated as positive signals minus negative signals.

Company	Positive	Negative	Net Score	Classification
DUOL	5	0	+5	Strong Candidate
SHAK	5	0	+5	Strong Candidate
BOOT	3	2	+1	Watchlist
ETSY	1	4	-3	Weak Candidate
CROX	0	5	-5	Weak Candidate

11. Earnings vs Cash-Flow Diagnostic

Company	Flag	Net Margin	CFO Margin
DUOL	Aligned positive	39.91%	37.38%
SHAK	Divergent periods present	3.16%	15.38%
BOOT	Divergent periods present	9.47%	7.72%
ETSY	Divergent periods present	5.65%	24.05%
CROX	Aligned positive	-2.01%	17.58%

The earnings/cash-flow flag is a diagnostic rather than an automatic investment judgment. It is intended to identify periods where earnings and operating cash-flow behavior deserve additional investigation.

12. SQL Analysis Completed

- Compared companies by positive and negative signal counts.
- Calculated net-direction score.
- Compared revenue growth and profitability.
- Compared operating, net and cash-flow margins.
- Analyzed operating-income dollar change.
- Reviewed improving and declining financial indicators.
- Reviewed earnings versus operating-cash-flow consistency.
- Created candidate classifications.
- Produced a final SQL-ready company-level screening output.

13. Final SQL Screening Result

The final SQL output combines candidate classification with the most important supporting financial evidence. The resulting ranking is:

Rank Group	Company	Classification	Net Score	Key Interpretation
1	DUOL	Strong Candidate	+5	Strong growth,

				profitability and cash-generation direction
1	SHAK	Strong Candidate	+5	Positive direction across all five screening signals
3	BOOT	Watchlist	+1	Positive growth but mixed profitability direction
4	ETSY	Weak Candidate	-3	Growth remains positive but several profitability/cash signals weakened
5	CROX	Weak Candidate	-5	Broad negative directional pattern under current framework

14. Financial Screening Interpretation

DUOL emerged as the strongest financial screening candidate. It achieved 95.36% revenue growth from FY2023 to FY2025, a 13.07% operating margin, 39.91% net margin, 37.38% CFO margin and a $148.83M increase in operating income.

SHAK also achieved a +5 net-direction score with 5 positive and 0 negative signals. Its FY2023–FY2025 revenue growth was 32.90%, with positive directional movement across operating margin, net margin, CFO margin and operating income.

BOOT is a Watchlist candidate because growth and cash-generation indicators are positive, while operating and net margins have declined.

ETSY and CROX are classified as Weak Candidates under the current financial-direction framework. This does not mean they are universally unattractive investments; it means their current financial signals are weaker under the defined screening rules.

15. MySQL Implementation

Database: investment_intelligence. Main screening table: investment_screening.

The final screening DataFrame contains 5 rows and 20 analytical columns. The dataset was successfully loaded into MySQL using a SQLAlchemy engine with the PyMySQL driver.

The direct pandas-to-SQL attempt using a raw PyMySQL connection produced a compatibility error. The workflow was corrected by using SQLAlchemy, which is the appropriate connection layer for pandas to_sql.

A second table, financial_data (company-year grain, 13 columns, key = ticker + fiscal_year), was created in the same database because Power BI trend charts need fiscal_year and yearly values, which the company-level screening table does not contain. The expected content is 15 rows (5 companies × FY2023–FY2025), with Shake Shack gross profit and Crocs operating cash flow stored as NULL because they are not available, not as zero.

The MySQL Workbench Table Data Import Wizard loaded only 6 of the 15 rows. The load was replaced with an explicit SQL INSERT script (TRUNCATE followed by 15 rows with NULLs). After running it, confirm COUNT(*) = 15 and the NULL checks before loading the table into Power BI.

16. Validation & QA

- Final screening DataFrame shape: 5 × 20.
- Five unique companies present.
- Expected candidate classifications produced.
- Positive signal count across companies: 14.
- Negative signal count across companies: 11.
- MySQL investment_screening table contains 5 company records.
- SQL outputs were cross-checked against the Python screening results.
- Power BI reconciliation: 5 companies; Strong 2, Watchlist 1, Weak 2; total positive/negative signals 14 / 11; ranking table and net-score chart match the SQL results.
- Rank measure must be re-verified after the tie-handling fix (expected ranks 1, 1, 3, 4, 5).

17. Error & Resolution Log

Issue	Cause	Resolution	Interview Lesson
Etsy duplicate _x/_y columns	Repeated merges created duplicate metric names.	Inspected columns and retained the validated metric fields.	Always inspect merge outputs before downstream calculations.
revenue_df missing after kernel restart	Jupyter kernel reset cleared in-memory variables.	Rebuilt the missing upstream object and reran dependent cells.	Notebook state must not be treated as permanent storage.
Boot Barn SEC 404	Initial CIK was incorrect.	Corrected the SEC identifier and reran extraction.	Validate regulatory identifiers before automation.
investment_intelligence_flags KeyError	The expected column was not present during partial notebook recovery.	Inspected the actual columns and reran the appropriate upstream logic.	Debug dependencies from actual object state, not assumptions.
pandas to_sql / PyMySQL error	Raw DBAPI connection was not supported cleanly by the installed pandas workflow.	Moved to a SQLAlchemy engine.	Use the correct abstraction layer for pandas database operations.
SQLAlchemy host parsing error	Connection string was malformed because of a special character in the credential.	Corrected/encoded the connection string.	Credentials containing URL-reserved characters must be encoded.
Large notebook state loss	Kernel restart removed all variables from memory.	Ran the complete notebook from a clean state and fixed errors sequentially.	A production-style pipeline should be reproducible from a clean environment.
Power BI: Cannot find table investment_screening	Power BI imported the table as "investment_intelligence investment_screening" (schema + table name).	Used the full table name in DAX, or renamed the table in the Data pane.	Check the exact object names Power BI creates on import before writing DAX.
Power BI: classification column not found	The MySQL table holds direction counts and flags but no classification column; the classification rule lived in the SQL query output.	Created Net Score and Classification calculated columns in DAX (Strong ≥ +3, Watchlist 0 to +2, Weak < 0 — thresholds inferred from current	Materialize business rules such as classification in the database so every tool uses the same rule.

		results and to be aligned with the SQL rule).	
DAX syntax error: "Total" is incorrect	Two measures were pasted into one formula bar.	Created one measure per New measure step.	Each DAX measure is created and saved separately.
Strong Candidates card showed 3 instead of 2	The measure condition did not isolate Strong Candidates; the KPI cards summed to 6 for 5 companies.	Corrected the measure so the cards reconcile to 5 / 2 / 1 / 2.	Reconcile KPI totals against known results before trusting a visual.
Rank column showed 1, 1, 1, 4, 5	Tie handling gave BOOT the same rank as the tied leaders.	Replaced the measure with "number of companies with a higher Net Score + 1" (expected 1, 1, 3, 4, 5); re-verify after applying.	Test ranking logic with ties.
Field parameter: syntax error on closing brace	A full formula was pasted over the existing one, leaving a duplicate closing brace.	Removed the extra brace.	Edit only the part that changes in generated DAX.
Metric field parameter gave a blank chart	Likely caused by manually editing the field-parameter definition; a normal column in the same chart displayed correctly, so the chart itself was fine.	Replaced it with a Metric Options table (DATATABLE) plus a SWITCH measure, Selected Metric Value, with Order as the sort column.	Isolate a fault by swapping one field at a time; prefer a simple, controllable design.
Company-year table missing in MySQL	Only investment_screening existed; trend charts need ticker, fiscal_year and yearly values.	Created financial_data (13 columns) from the Financial Data sheet.	Confirm the data grain supports every planned visual before building.
Import Wizard loaded 6 of 15 rows	The wizard stopped part-way (likely on rows with blank numeric cells; exact cause not confirmed).	Used a SQL INSERT script with explicit NULLs; verify COUNT(*) = 15.	Prefer scripted loads for small datasets: explicit and repeatable.
Metric buttons and company bars in changing order	Text slicers sort alphabetically, and the chart re-sorts by value whenever the metric changes.	Sort by column (Order) for the buttons; a Company Order calculated column (RANKX on Net Score, then revenue growth) is used to fix the Strong → Weak order.	Control sort order explicitly with a numeric helper column.

18. End-to-End Architecture

SEC EDGAR/XBRL → Python extraction → reusable ETL → annual filtering/deduplication → standardization → company-year financial dataset → derived metrics & QA → investment intelligence → final screening dataset → MySQL → SQL analysis → candidate classification → Power BI.

19. Current Limitations

- The current model is financial-first and is not complete investment due diligence.
- The directional score is transparent but not a valuation model.

- Valuation, TAM, competition, strategic fit, management quality and industry outlook are not yet included.
- The current screening table is a company-level analytical output.
- The Power BI dashboard is partly built: Executive Overview is complete; Performance Comparison is in progress; Company Detail and Method & Limits are still to be built.
- Final investment conclusions must distinguish screening evidence from broader investment judgment.
- The classification is currently derived in Power BI with DAX because the MySQL screening table has no classification column; the thresholds should be aligned with the SQL classification logic and stored in the database.
- Two items to review in the notebook/SQL: the Crocs earnings-vs-cash-flow flag reads "Aligned positive" although net margin is −2.01%, and Crocs shows +1.98% revenue growth over FY2023–FY2025 but zero positive signals (the growth signal may use the latest year).

20. Power BI Dashboard — Build Log & Next Steps

20.1 Design approach

- Three HTML prototypes were reviewed before building: an analytics-style tabbed layout, a more creative "Deal Room" layout, and a simple single-page layout. The simple version was chosen because it is the easiest to read and maps directly to standard Power BI visuals (cards, bar charts, slicers, tables).
- Power BI Desktop is connected to the MySQL database investment_intelligence; the dashboard is clearly labelled as financial screening and decision support.

20.2 Page status

Page	Content	Status
Executive Overview	Title banner; KPI cards (companies screened, Strong, Watchlist, Weak, positive/negative signals); net-score chart coloured by classification; candidate ranking table; "Reading the result" note	Completed (confirm Rank column shows 1, 1, 3, 4, 5)
Performance Comparison	Metric slicer with five buttons and one switching chart (working); fixed Strong → Weak company order and classification colours; operating-income-change chart; full comparison table	In progress
Company Detail	Company slicer, five KPI cards, revenue chart, margin lines, interpretation text; needs financial_data loaded and related to the screening table on ticker	Planned
Method & Limits	Business problem, architecture, scoring method and limitations	Planned

20.3 Data model & DAX objects

Object	Type	Purpose
Net Score	Calculated column	Positive minus negative direction count
Classification	Calculated column	Strong / Watchlist / Weak from Net Score (thresholds to be aligned with SQL rule)
Classification Color	Measure	Hex colour per classification for bars and table cells
Companies Screened, Strong Candidates, Watchlist, Weak Candidates	Measures	KPI cards (distinct company counts)
Total Positive Signals, Total Negative Signals, Positive / Negative Signals	Measures	Signal totals and the "14 / 11" card
Rank	Measure	Companies with a higher Net Score + 1; ties share a

		rank
Metric Options	Calculated table	Names and order of the metric slicer buttons
Selected Metric Value	Measure	SWITCH over the selected metric button
Company Order	Calculated column	Fixes the company order Strong → Weak on charts

20.4 Next steps

- Finish Performance Comparison: Company Order sort, classification colours, data labels, dynamic title, operating-income-change chart, comparison table.
- Run the SQL INSERT script, confirm 15 rows and the NULL checks, then load financial_data into Power BI and relate it to the screening table on ticker.
- Build Company Detail using Revenue $M and margin measures that return blank (not zero) when gross profit or operating cash flow is NULL.
- Add a page navigator, build Method & Limits, and align the Classification thresholds with the SQL rule.
- Capture dashboard screenshots and update the portfolio narrative and interview story.

21. Interview-Ready Project Explanation

I built a source-backed corporate venture screening workflow using SEC EDGAR/XBRL data for five public companies across FY2023–FY2025. I first validated the extraction approach on Etsy and then generalized it into reusable Python ETL with annual filtering, deduplication, standardization and QA. I derived growth, profitability and cash-flow indicators and converted their direction into transparent screening signals. I then persisted the final analytical dataset in MySQL and used SQL to compare companies, calculate a net-direction score and classify them into Strong Candidate, Watchlist and Weak Candidate groups. Duolingo and Shake Shack emerged as the strongest financial-screening candidates. The next stage is to visualize the validated analytical layer in Power BI and then extend the project with broader investment factors.

In Power BI I connected to the MySQL screening data, agreed the layout through prototypes, and built a simple executive page with KPI cards, a classification-coloured net-score chart and a ranking table using DAX. I diagnosed and documented the issues along the way — table naming, a missing classification column, tie handling in ranking, a field parameter that returned a blank chart, and an import that loaded only 6 of 15 rows — and fixed each one. The remaining work is the company-level detail page and the final portfolio narrative.

22. Documentation Maintenance

- This is Version 1.5 of the living project documentation.
- Future work should be added to the relevant existing section instead of creating disconnected notes.
- Update the roadmap whenever a phase changes status.
- Record meaningful technical errors and resolutions in the Error & Resolution Log.
- Keep financial screening conclusions separate from broader investment recommendations.
- Record Power BI and DAX issues and resolutions in the same Error & Resolution Log.
- After Power BI is completed, update the dashboard architecture, insights and final portfolio narrative.

23. Current Project Checkpoint

Checkpoint reached: Python notebook runs end-to-end; financial screening is validated; MySQL load and SQL analysis are complete; Power BI is connected to MySQL and the Executive Overview page is built and reconciled to the SQL results. A company-year table (financial_data) has been created in MySQL and its load is to be confirmed before the Company Detail page is built. Performance Comparison is in progress.